Homework 3: Perceptual Metric Reproduction & Failure Analysis

115-1 Human Perceptual-Centric Vision and Multimodal Representation Learning

Assignment Schedule

Released: October 27, 2026 (Week 8); **Related Lecture:** October 27, 2026 (Week 8) — Perceptual Metrics and Image/Video Quality Assessment; **Due:** November 17, 2026 (Week 11), before class. Students will have approximately **three weeks** to complete the assignment.

1 Objective and Learning Goals

The goal of this assignment is to understand how perceptual quality metrics are evaluated against human judgments and, more importantly, why different metrics fail under different conditions.

Students will reproduce several representative image-quality metrics, compare their predictions with subjective quality scores, and identify cases where traditional and learned perceptual metrics disagree with humans or with one another.

The intended progression is:

Run the Metric → Compare with Human Judgments → Find Disagreements → Explain the Failure

The emphasis is not simply on obtaining the highest correlation. Students should learn that different metrics encode different assumptions about what visual information matters to human perception.

2 Task

This homework should be completed in teams of **2–3 students**.

Each team will evaluate at least **three perceptual quality metrics** on a public image-quality dataset containing human subjective quality scores.

The comparison must include:

- at least one traditional fidelity or structural metric (**PSNR or SSIM**);
- at least one learned perceptual metric (**LPIPS or DISTS**); and
- at least one additional metric chosen by the team.

Possible additional metrics include PSNR, SSIM, MS-SSIM, LPIPS, DISTS, or another relevant perceptual metric discussed in class.

Students may use official implementations or existing open-source packages. Reimplementing the complete metric from scratch is **not required**. However, students must understand what each metric measures and verify that the implementation is used correctly.

3 Dataset

Use a public IQA dataset containing distorted images and corresponding subjective human quality scores (MOS or DMOS).

Possible datasets include:

- LIVE IQA;
- TID2013;
- KADID-10k;
- PIPAL; or
- another appropriate dataset approved by the instructor.

Using the entire dataset is not required if computational cost is high. A representative subset may be used, provided that the sampling procedure and included distortion types are clearly described.

4 Minimum Requirements

For each evaluated metric, compute its prediction for the selected images and compare the predictions against the subjective human quality scores.

Report at least:

SRCC (Spearman Rank Correlation)

and

PLCC (Pearson Linear Correlation).

Before computing PLCC, students should verify whether the metric direction is consistent with the subjective score direction (e.g., higher-is-better versus lower-is-better).

The evaluation should include multiple distortion types whenever possible. Students should not rely only on the overall dataset correlation; they should also examine whether metric performance changes across different distortion categories.

5 Failure-Case Analysis

The most important part of this assignment is the **failure-case analysis**.

Identify at least **5 representative images or image pairs** where:

- two metrics strongly disagree;
- a metric prediction strongly disagrees with the human quality score; or
- the ranking produced by a traditional metric differs substantially from that produced by a learned perceptual metric.

For each selected case, visually inspect the distortion and propose a perceptual explanation for the disagreement.

Possible factors include structure, texture, spatial localization, contrast, color, blur, noise, semantic content, salient regions, or sensitivity to small geometric changes.

The goal is to answer:

What visual information does this metric emphasize or ignore?

6 Required Content

The report should contain the following components.

1. Metrics and Dataset Briefly introduce the selected dataset, distortion types, subjective scores, and the evaluated metrics. Explain the main perceptual assumption behind each metric.

2. Experimental Setup Describe dataset selection, preprocessing, metric implementations, score direction, and evaluation procedure. The description should contain enough information for another student to approximately reproduce the experiment.

3. Quantitative Results Report the SRCC and PLCC of each metric using a table or figure. When possible, also compare performance across different distortion categories.

Students should interpret the results rather than simply state which metric obtains the highest score.

4. Failure Cases Present at least **5 representative failure or disagreement cases**. For each case, show the relevant image(s), metric predictions, and human subjective score or ranking.

Explain why the metrics may behave differently from one another or from human perception.

5. Discussion Summarize what the experiment reveals about the strengths and limitations of the evaluated metrics.

Discuss whether any single metric consistently reflects human perception and identify at least one perceptual factor that is not adequately captured by the evaluated methods.

7 Practical Notes

Do not spend excessive time reimplementing existing metrics. Official repositories and established packages may be used. The purpose of this homework is metric evaluation and perceptual analysis rather than software engineering.

Check score direction carefully. For example, PSNR and SSIM generally increase with quality, while some distance-based metrics increase with perceptual difference. Incorrect score direction can produce misleading correlations.

Do not cherry-pick only visually dramatic examples. Failure cases should be selected based on quantitative disagreement first, followed by visual interpretation.

8 Submission

Submit:

(1) Report: A concise PDF describing the experiment, quantitative results, failure cases, and interpretation.

(2) Code: Scripts or notebooks required to reproduce the metric evaluation and figures.

The recommended report length is **3 pages**, with an absolute maximum of **4 pages**, excluding references and the Contribution Statement.

Include a short **Contribution Statement** describing each team member's role.

All submissions must follow the course-wide *Academic Integrity and Generative AI Policy*.

9 Grading

Component	**Points**
Correct metric implementation and understanding	15
Experimental setup and reproducibility	15
Quantitative evaluation (SRCC / PLCC)	20
Failure-case selection and visualization	20
Perceptual explanation and critical analysis	20
Clarity, organization, and presentation	10
Total	**100**

A strong submission should go beyond the conclusion that one metric has a higher correlation than another. The analysis should explain **when and why** different metrics agree or disagree with human perception.